SNP- and INDEL-specific filtering of variant calls

Raw variant calls from GATK HaplotypeCaller

Separate SNPs and INDELs
because variant classes have different error profiles

SNP filtering thresholds

- QD ≥ 1.8
- FS ≤ 60
- SOR ≤ 3
- MQ ≥ 40

MQRankSum ≥ −12.5
ReadPosRankSum ≥ −8.0

Purpose: remove SNPs with poor quality-by-depth, strand bias, weak mapping support, or read-position bias

INDEL filtering thresholds

- QD ≥ 1.8
- FS ≤ 200
- SOR ≤ 10

ReadPosRankSum ≥ −20.0
More relaxed thresholds used
for INDEL-specific calling sensitivity

Rationale: INDEL calls are more affected by local alignment and sequencing-context errors than SNPs

Filtered high-confidence SNP and INDEL datasets
used for variant-effect prediction and downstream interpretation